6 **(a)** State Faraday's law of electromagnetic induction.

.....

.....

..... [1]

(b) An aircraft is flying horizontally at constant speed *v* through the Earth's magnetic field, as shown in Fig. 6.1.

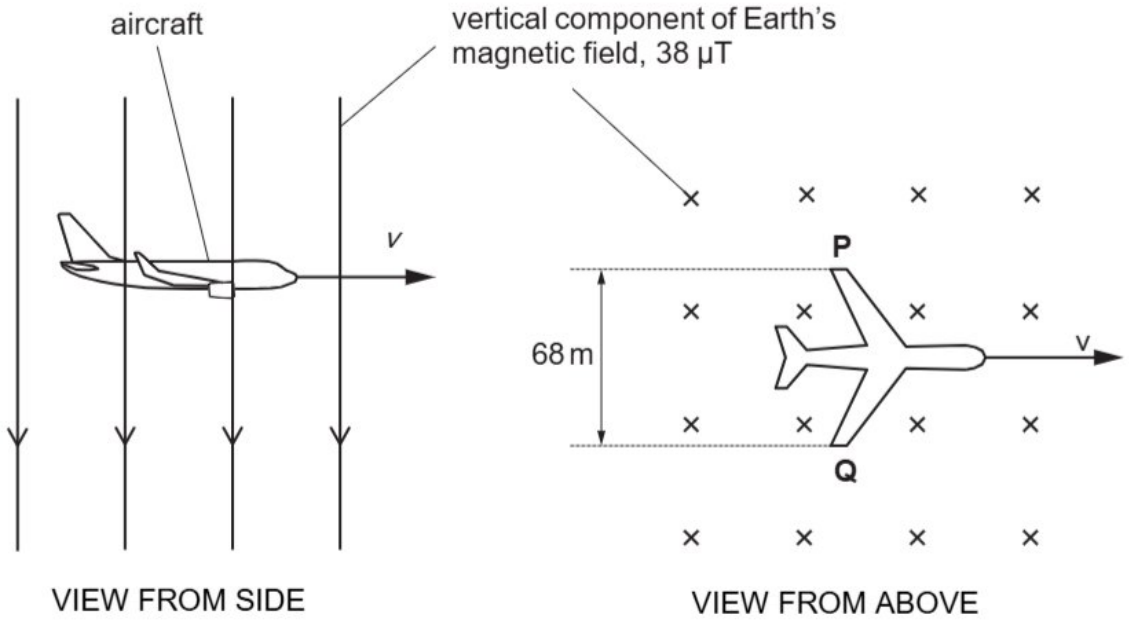


Fig. 6.1

At the location of the aircraft, the vertical component of the Earth's magnetic field is 38 μT towards the ground.

The distance between the wingtips P and Q of the aircraft is 68 m.

As the aircraft moves through the magnetic field, an electromotive force (e.m.f.) of 0.55 V is induced between the wingtips P and Q.

(i) Calculate the magnetic flux cut by the wings of the aircraft in a time of 14 s.

Provide a suitable unit with your answer.

magnetic flux = ...

unit = ... [2]

(ii) Determine the area of flux cut by the wings in a time of 14 s.

area = ... m^2 [1]

(iii) Use your answer in **(b)(ii)** or otherwise to determine the speed v of the aircraft.

v = ... $m\ s^{-1}$ [2]

(iv) Using Fleming's Left Hand Rule, explain clearly why the wingtip P is at the higher potential compared to wingtip Q.

.....

.....

.....

.....

..... [2]

(c) A sinusoidal power source with alternating voltage V varies with time t according to

$$V = 16\cos(38\pi t)$$

where V is in V and t is in s.

The circuit connection with the two resistors R_1 and R_2 each having resistance R and an ideal diode is shown in Fig. 6.2

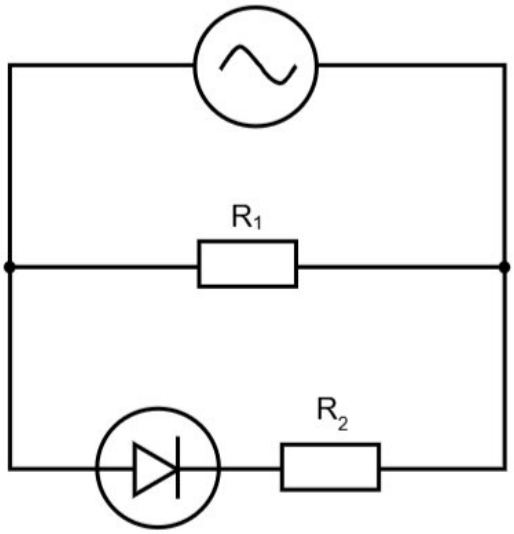


Fig. 6.2

(i) Fill in the missing values in Table 6.1.

resistor	r.m.s. voltage, $V_{r.m.s.}$ / V	peak voltage, V_0 / V
R_1		
R_2		

Table 6.1

[1]

(ii) Compare the r.m.s voltages and peak voltages across resistors R_1 and R_2, and explain your answers.

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.....

.....

.....

.....

..... [2]

[Total: 11]